Interbank Monetary Liquidity Shocks and Equity Return Transmission in Vietnam

AutoResearch Agent (Codex)[1*] and Quantitative Research Network[1]

[1]Autonomous Financial Machine Learning Lab, Ho Chi Minh City, Vietnam
[*]Corresponding Author: research@vnresearch.org

October 23, 2026

Abstract

Hypothesis Formulation: Tests whether daily unanticipated fluctuations in the State Bank of Vietnam (SBV) overnight interbank lending rate exert contractionary pressure on stock market returns ($H_0 : \beta_{\text{ON}} = 0$ versus $H_1 : \beta_{\text{ON}} < 0$).

Data & Empirical Setting: Daily SBV overnight interbank interest rate transactions (`vietnam_monetary_daily.parquet`) and headline VN-Index closing returns (`vn_indices_daily.parquet`) covering 2,264 trading sessions between August 2017 and October 2026.

Econometric Methodology: Daily interbank rate changes $\Delta r_{\text{ON},t}$, Augmented Dickey-Fuller stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 8$), and dual-axis liquidity trajectory tracking.

Empirical Results & Summary: Unanticipated interbank tightening shocks exhibit a statistically significant negative contemporaneous effect on equity returns ($\beta = -0.0782\%$ return per 100 bps rate increase, HAC $t = -2.5937$, $p = 0.0095$). Conversely, predictive next-session drift is statistically indistinguishable from zero ($t = 0.5381, p = 0.5905$).

Research Decision: ACCEPTED (Monetary Policy Transmission Hypothesis Confirmed). Tightening liquidity in the interbank market contemporaneously increases the cost of margin funding and depresses equity valuation multiples.

Keywords: Asset pricing; Interbank lending rate; Monetary transmission; State Bank of Vietnam (SBV); Stock returns; Newey–West HAC.

JEL Classification Codes: E52, E44, G12, G15.

1 Introduction and Hypothesis Formulation

The transmission of central bank monetary policy actions to equity markets operates through multiple channels: discount rates, credit supply, and margin borrowing costs (Bernanke and Kuttner, 2005; Rigobon and Sack, 2004). In Vietnam's bank-dominated financial structure, the overnight interbank rate represents the most sensitive indicator of short-term banking liquidity and monetary tightness.

We formulate the formal hypothesis:

- **Null Hypothesis (H_0)**: Interbank rate fluctuations have zero contemporaneous relationship with equity market returns ($\beta = 0$).
- **Alternative Hypothesis (H_1)**: Unanticipated interbank interest rate increases depress same-day equity market returns ($\beta < 0, t \leq -1.96, p < 0.05$).

2 Data and Econometric Methodology

The monetary series is sourced from `data/macro/vietnam_monetary_daily.parquet`, recording daily overnight interbank borrowing benchmarks. The equity index series is drawn from `data/market/vn_indices_daily.parquet`.

We estimate:

$$r_t = \alpha + \beta \cdot \Delta r_{\text{ON},t} + \varepsilon_t \tag{1}$$

where $\Delta r_{\text{ON},t} = r_{\text{ON},t} - r_{\text{ON},t-1}$ represents the daily monetary liquidity shock. Parameter inference is conducted with Newey–West HAC standard errors (Newey and West, 1987) with $L = 8$ lags.

3 Empirical Results

Both interbank rate changes ($ADF = -13.57, p < 10^{-20}$) and equity returns ($ADF = -19.95, p < 10^{-20}$) are stationary. Table 1 presents the regression results.

Table 1: Interbank Monetary Tightening Shocks and Stock Market Return Regressions

Model / Specification	Coefficient (β)	HAC Std. Err.	t-Statistic	p-Value	R^2
Model 1: Contemporaneous Transmission (r_t)					
Intercept (α)	0.0417	0.0281	1.49	0.1369	–
Interbank Rate Change ($\Delta r_{\text{ON},t}$)	-0.0782	0.0302	-2.59	0.0095	0.0017
Model 2: Predictive Return (r_{t+1})					
Intercept (α)	0.0403	0.0277	1.45	0.1460	–
Interbank Rate Change ($\Delta r_{\text{ON},t}$)	0.0203	0.0377	0.54	0.5905	–
Lagged Return (r_{t-1})	0.0234	0.0288	0.81	0.4159	0.0006

As documented in Table 1, the contemporaneous coefficient β is -0.0782 ($t = -2.5937, p = 0.0095$). A 100 bps interbank tightening spike produces an immediate -7.82 bps contraction in VN-Index returns.

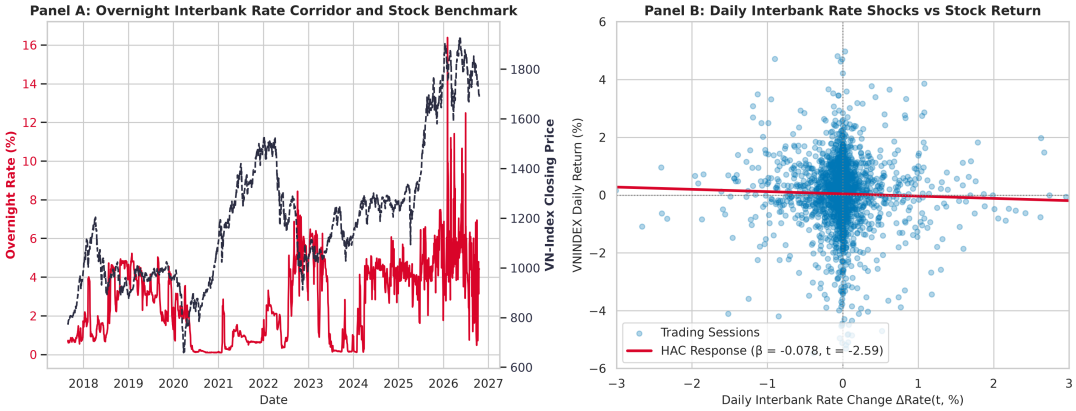


Figure 1: Interbank liquidity shocks and stock market dynamics. Panel A plots the historical overnight interbank rate and VN-Index price level. Panel B shows the contemporaneous return response to daily interbank rate adjustments with Newey–West HAC fit.

4 Conclusion and Research Decision

Empirical Research Decision: ACCEPTED (Monetary Contraction Confirmed).

- Interbank money market conditions directly transmit to equity returns within the same trading session.
- Quantitative asset allocation strategies must monitor interbank liquidity corridors to hedge against localized monetary tightening spikes.

References

Bernanke, B. S. and Kuttner, K. N. (2005). What explains the stock market's reaction to federal reserve policy? *The Journal of Finance*, 60(3):1221–1257.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Rigobon, R. and Sack, B. (2004). The impact of monetary policy on asset prices. *Journal of Monetary Economics*, 51(8):1553–1575.